

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

COMPUTER VISION [CSE-4031]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply median filter on the pixels (1,3), (4,1) and (2,4) of the image matrix given below. Write the output matrix. Assume zero padding near the border.

A)

3	1	100	2
0	6	9	75
1	4	4	15
210	8	11	10

(4)

B)

Apply bit-plane slicing on the given image below. What is the application of this technique in image processing?

6	7	6	6	7
0	0	0	1	2
1	1	1	2	3
4	5	5	4	2
6	6	6	7	7

(3)

C)

Let $(R, G, B) = (0.582, 0.1608, 0.1227)$. Convert this RGB color model to HSI color model. Illustrate this computation with appropriate mathematical equations and clear steps.

(3)

- 2) How Laplacian filter technique can be used to detect edges in the image? Explain with appropriate masks and mathematical equations to detect edges in x and y-direction.

(5)

A)

B)

Explain with appropriate steps how the Hough transform can be used to detect lines in the image. What are its limitations and drawbacks in line feature detection?

(5)

- 3) Two cameras are arranged arbitrarily to take the image of a scene. Write stereo rectification procedure to make image planes of these cameras parallel to each other.

(5)

A)

B)

How do you determine extrinsic parameters in camera calibration process? Explain with

(5)

appropriate projection diagram and equations.

- 4) How does object tracking differ from motion estimation? Explain the need for object tracking in a sports video? (3)
- A)
- B) Develop Kanade-Lucas-Tomasi (KLT) algorithm to track objects in video based on optical flow. How this technique differs from translational alignment? (4)
- C) Explain any two applications of object tracking in video. Provide an analysis of the motion tracking method that can be used in these applications. (3)
- 5) Explain with a neat diagram various steps involved in searching similar images in the image database in response to query image. (5)
- A)
- B) The hospital requires computer-aided diagnosis system to assist physicians in diagnosing the disease. Explain how computer vision steps can be applied in CAD systems starting from image acquisition to disease recognition. (5)

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